

LeetCode Solution: 1894. Find the Student that Will Replace the Chalk

Problem Overview

The problem asks us to determine which student will be the first to run out of chalk given an initial amount k and a sequence of chalk usage per student. Since students consume chalk in a cyclical manner, a naive simulation would be inefficient if k is very large.

Logic & Approach

- **Cycle Optimization:** Calculate the sum of the entire chalk array. This represents one full round of consumption.
- **Modulo Operator:** Instead of simulating thousands of rounds, we compute $\text{rem} = k \% \text{sum}$. This skips all full cycles and leaves us with the chalk remaining at the start of the final partial cycle.
- **Simulation:** Iterate through the array one last time. If the student's chalk requirement is greater than our rem , that student is the one who will run out. Otherwise, subtract their requirement from rem and continue.

Complexity

Time Complexity: $O(N)$, where N is the number of students. We traverse the array to compute the sum and once more to find the culprit.

Space Complexity: $O(1)$, as we only use a few variables for storage regardless of input size.

Solution Code

```
class Solution {
public:
    int chalkReplacer(vector& chalk, int k) {
        int n=chalk.size();
        long long sum=accumulate(chalk.begin(),chalk.end(),0LL);

        long long rem=k%sum;
```

```
    for(int i=0;i<rem){  
        return i;  
    }  
  
    rem-=chalk[i];  
}  
  
return -1;  
}  
};
```

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